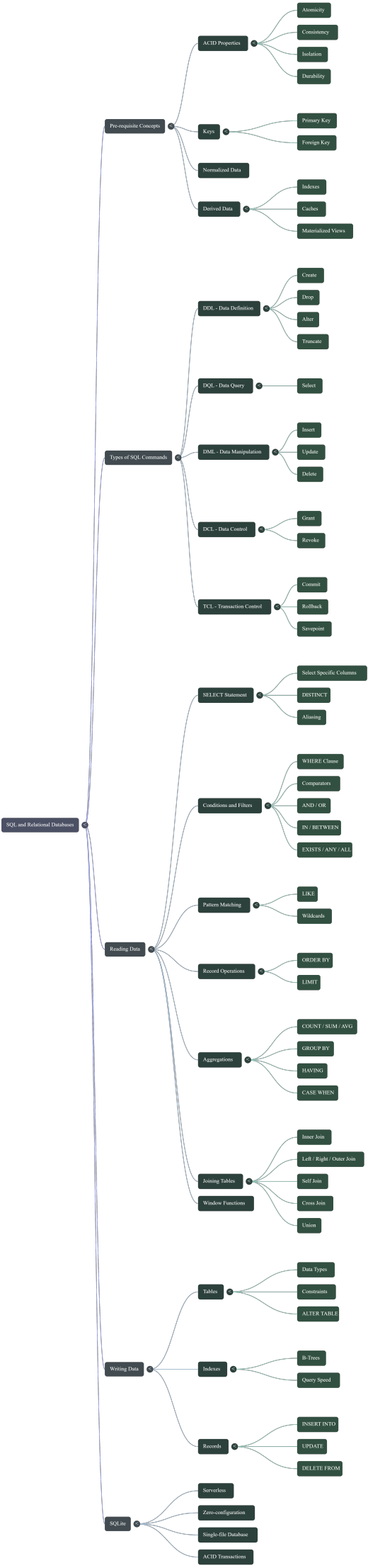

SQL and Relational Databases
Pre-requisite Concepts
ACID Properties
Atomicity
Consistency
Isolation
Durability
Keys
Primary Key
Foreign Key
Normalized Data
Derived Data
Indexes
Caches
Materialized Views
Types of SQL Commands
DDL - Data Definition
Create
Drop
Alter
Truncate
DQL - Data Query
Select
DML - Data Manipulation
Insert
Update
Delete
DCL - Data Control
Grant
Revoke
TCL - Transaction Control
Commit
Rollback
Savepoint
Reading Data
SELECT Statement
Select Specific Columns
DISTINCT
Aliasing
Conditions and Filters
WHERE Clause
Comparators
AND / OR
IN / BETWEEN
EXISTS / ANY / ALL
Pattern Matching
LIKE
Wildcards
Record Operations
ORDER BY
LIMIT
Aggregations
COUNT / SUM / AVG
GROUP BY
HAVING
CASE WHEN
Joining Tables
Inner Join
Left / Right / Outer Join
Self Join
Cross Join
Union
Window Functions
Writing Data
Tables
Data Types
Constraints
ALTER TABLE
Indexes
B-Trees
Query Speed
Records
INSERT INTO
UPDATE
DELETE FROM
SQLite
Serverless
Zero-configuration
Single-file Database
ACID Transactions